

Mixture Score-Based Diffusion Models with Multiple Choice Learning

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Abstract

Score-based diffusion models have emerged as a powerful class of generative models, yet a single denoising network must implicitly represent all modes of the data distribution, which can limit both sample quality and diversity. This report explores the combination of score-based diffusion with *Multiple Choice Learning* (MCL), a structured prediction framework that trains K competing expert networks with a stochastic Winner-Takes-All (WTA) loss. The methodology is applied to MNIST with $K = 4$ experts, with four collapse-mitigation training variants (hard, annealed, relaxed, and resilient WTA), and a study of five inference-time routing strategies: a single expert over the full trajectory, per-step random and best-norm heuristics, score mixture averaging, and a learned gating network. A dedicated cross-variant ablation on an A100-80GB GPU is conducted and shows that the training objective fundamentally shapes inference-time behaviour: *hard* and *resilient* WTA produce a single dominant expert (yielding a best FID of 13.49 with gating at $K = 4$, but an effective ensemble size of one; pushing K to 8 under resilient MCL further lowers this to the global best of 13.00 via hidden-expert routing), while *annealed* and *relaxed* WTA produce genuinely multi-skilled experts (gating reaches FID = 88.25, *strictly beating* the monolithic baseline at 111.44, with Precision 0.644 and Recall 0.836). A rigorous distinction between inter-class and intra-class diversity is provided, along with a formalisation of the resulting quality-versus-true-mixture trade-off; the methodologically-correct reading of MCL+diffusion is argued to be the annealed-WTA regime, in which the gating network provides a non-trivial multi-expert improvement over both the baseline and any single expert.

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1 Introduction

Diffusion models [1, 2] learn to reverse a gradual noising process, transforming simple Gaussian noise into samples from a complex data distribution. In the continuous-time variance-exploding (VE) formulation, one trains a neural network $\epsilon_\theta(\mathbf{x}_t, \sigma)$ to predict the noise component of a corrupted sample, and new data is generated by solving a deterministic probability-flow ordinary differential equation (ODE).

While a single model can in principle capture the entire data distribution, in practice its finite capacity forces compromises: modes may be under-represented, and the model may average between distinct structures instead of committing to one. *Multiple Choice Learning* (MCL) [3, 4] addresses this by training K parallel expert networks under a Winner-Takes-All (WTA) objective that encourages specialization. Each expert becomes responsible for a distinct region of the output space, forming a Voronoi-like partition.

The central question of this project is: **how should one route between specialized experts at inference time, and how does the training objective shape the set of meaningful routing strategies?** Five strategies of increasing sophistication are implemented and compared (single expert, per-step random, per-step best-norm, score mixture, learned gating) and their effects are analyzed on the quality–diversity trade-off. A second question, which emerges empirically from the experiments in this report, is **whether the WTA training objective produces a genuine mixture of experts at all:** the experiments show that this depends sharply on the collapse-mitigation strategy used during training, and that the standard reading of “MCL + learned gating beats the baseline” holds cleanly only in the regime where all experts acquire non-trivial competence.

Related Works. This report sits at the intersection of three lines of work — score-based generative modelling, Multiple Choice Learning, and mixture-of-experts inference — summarised below.

- **Score-based diffusion models.** Ho et al. [1] introduced the DDPM framework, training a neural network to denoise samples corrupted by a fixed Gaussian Markov chain. Song et al. [2] unified DDPM with score-matching into a continuous-time SDE formulation, and introduced the two parameterisations used here: the variance-exploding (VE) and variance-preserving (VP) SDEs, together with the deterministic probability-flow ODE that enables fewer-step sampling. Karras et al. [5] (EDM) studied the design space of noise-preconditioning, weighting, and sampler choices; the VE formulation adopted here follows with log-uniform σ -sampling during training and a geometric σ -grid at inference, together with the Heun second-order ODE integrator they recommend.
- **Multiple Choice Learning.** The WTA loss was introduced by Guzmán-Rivera, Batra, and Kohli [3] for structured prediction tasks where multiple plausible outputs exist for a single input. Lee et al. [4] extended this to diverse deep ensembles with stochastic gradient descent, and documented the *hypothesis-collapse* failure mode that motivates all four training variants studied here. Perera, Letzelter et al. [8] recently proposed *Annealed MCL*, replacing the hard argmin with a temperature-scheduled softmax (Eq. (12)); Letzelter et al. [9] concurrently proposed *Resilient MCL*, which introduces per-expert learned scoring heads trained by cross-entropy against the true error-argmin (Eq. (15)). We are not aware of prior work combining MCL with score-based diffusion in this exact setup and systematically comparing the four training variants on a common benchmark.
- **Mixture-of-experts inference and gating.** Per-step expert selection in an ODE integrator is conceptually close to mixture-of-experts architectures [10], but this setting differs in three key ways: (i) experts are not conditioned on a learned gating *during training* — routing is implicit in the WTA loss; (ii) routing at inference time is applied *sequentially*

along an ODE trajectory, so coherence across steps is a separate concern from per-step competence; (iii) the gating network trained in Section 3.3.3 is a *post-hoc* classifier over winner labels produced by the trained experts, rather than a jointly-trained gating component. The Pareto analysis in Section 4.3 shows that these three distinctions are not academic: strategies that would be standard in MoE (uniform or score-weighted mixing) are catastrophic in the diffusion-MCL setting when WTA training has produced true mode-exclusive specialists.

- **Evaluation metrics.** Evaluation relies on Fréchet Inception Distance (FID) [6] and the Improved Precision/Recall metric of Kynkäänniemi et al. [7]. Because the standard Inception-v3 features are poorly calibrated for 28×28 grayscale digits, a small CNN classifier is used instead, trained on MNIST itself (Section 4.1); this choice is discussed in the caveat paragraph at the end of Section 4.2.

Summary.

- **Cross-variant ablation of MCL training objectives for score-based diffusion.** We implemented the full training and inference pipeline for four collapse-mitigation variants (hard, annealed, relaxed, and resilient WTA) and combined them with five inference routing strategies (single expert, per-step random, per-step best-norm, score mixture, and a learned gating network), yielding a systematic cross-tabulation of training objective against inference rule under FID, Precision, and Recall.
- **Two qualitatively distinct regimes of MCL+diffusion.** We isolated a *strong-specialist* regime (hard, resilient), in which any per-step routing collapsed to catastrophic FID and the learned gating reduced to expert-indicator inference, from a *soft-specialist* regime (annealed, relaxed), in which all experts acquired overlapping competence and gating provided a genuinely multi-expert improvement over the monolithic baseline.
- **K and batch size as implicit anti-collapse regularisers.** We ran two orthogonal sweeps over $K \in \{2, 3, 4, 6, 8\}$ and $B \in \{128, 256, 512, 1024\}$ and observed a phase-transition-like jump between $K = 2$ and $K = 3$ under annealed WTA, a non-monotone gated-FID optimum at $K = 6$, and a sharp shift from $B = 256$ to $B = 1024$ where the gated FID improved from 131 to **25.74** (all in this single-seed protocol).
- **Hidden-expert routing at resilient $K = 8$.** We showed that, at $K = 8$ under resilient MCL, the default expert was dead (single-expert FID = 3206) yet the learned gating network located a surviving expert among the eight and routed every ODE step to it, yielding the global best FID of the study (**13.00**) — an effectively single-expert operating point that establishes the upper bound on the quality axis.
- **Quality-versus-true-mixture trade-off.** We proposed a rigorous distinction between inter-class and intra-class diversity, from which we argued that the annealed-WTA configuration is the methodologically-correct instantiation of MCL+diffusion, while the hard and resilient results were reframed as upper-bound benchmarks obtained through expert collapse rather than through a principled mixture mechanism.

2 Background & Baseline Implementation

2.1 Score-Based Diffusion Models

Let p_{data} denote the data distribution on $\mathbb{R}^d$. Given a monotone schedule $\sigma : [0, T] \rightarrow \mathbb{R}_+$, a clean point $\mathbf{x}_0 \sim p_{\text{data}}$ is corrupted by adding independent Gaussian noise:

$$\mathbf{x}_t = \mathbf{x}_0 + \sigma(t) \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (1)$$

We write p_t for the distribution of $\mathbf{x}_t$. For $\sigma(T)$ large enough, p_T is approximately Gaussian, and the generative problem reduces to *reversing* this convolution: one starts from $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \sigma(T)^2 \mathbf{I})$ and integrates the probability-flow ODE

$$\frac{d\mathbf{x}}{dt} = -\dot{\sigma}(t) \sigma(t) \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \quad (2)$$

back to $t = 0$. The vector field is driven by the *score* $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$, which is intractable in closed form but admits a clean connection to the optimal denoiser. For the Gaussian kernel in (1), Tweedie’s formula gives

$$\mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t] = \mathbf{x}_t + \sigma(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t), \quad (3)$$

and rearranging in terms of the noise realisation $\boldsymbol{\epsilon} = (\mathbf{x}_t - \mathbf{x}_0)/\sigma(t)$ collapses the score to

$$\nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t) = -\frac{\boldsymbol{\epsilon}}{\sigma(t)}. \quad (4)$$

A network $\boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, \sigma)$ trained to predict $\boldsymbol{\epsilon}$ therefore recovers the score up to a σ -dependent rescaling, and the resulting *denoising score matching* objective

$$\mathcal{L}_{\text{DSM}}(\theta) = \mathbb{E}_{\mathbf{x}_0, \sigma, \boldsymbol{\epsilon}} \left[\left\| \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, \sigma) - \boldsymbol{\epsilon} \right\|_2^2 \right] \quad (5)$$

is what every variant in this report optimises — equivalent, up to a σ -dependent weighting, to standard score matching. In this report σ ranges from $\sigma_{\min} = 0.01$ to $\sigma_{\max} = 80$, spanning four orders of magnitude; training therefore samples $\log \sigma \sim \mathcal{U}(\log \sigma_{\min}, \log \sigma_{\max})$, so that gradients are balanced across scales rather than dominated by the low-noise regime.

2.2 Network Architecture

The noise predictor $\boldsymbol{\epsilon}_{\theta}$ is parameterized as a U-Net with residual blocks. The noise level σ is encoded via a sinusoidal positional embedding applied to $\log \sigma/4$, followed by a two-layer MLP producing a time embedding vector of dimension $d_t = 128$. The U-Net consists of an encoder–bottleneck–decoder structure with skip connections. Each residual block applies GroupNorm, 3×3 convolutions, and additive time conditioning. Downsampling and upsampling are performed via strided and nearest-neighbor convolutions, respectively. With a base channel width of 32 and channel multipliers (1, 2), the architecture is compact yet expressive enough for the 28×28 grayscale MNIST images.

An exponential moving average (EMA) of the model weights with decay $\beta = 0.999$ is maintained throughout training and used for all inference.

2.3 Deterministic Sampling via Probability Flow ODE

Sampling proceeds by solving the probability-flow ODE that transports the prior $\mathcal{N}(\mathbf{0}, \sigma_{\max}^2 \mathbf{I})$ to the data distribution. In the VE parameterization, the ODE takes the form

$$\frac{d\mathbf{x}}{d\sigma} = \boldsymbol{\epsilon}_{\theta}(\mathbf{x}, \sigma), \quad (6)$$

integrated from $\sigma = \sigma_{\max}$ down to $\sigma \approx 0$. The noise schedule for integration is constructed as a geometric progression:

$$\sigma_i = \sigma_{\max}^{1-r_i} \sigma_{\min}^{r_i}, \quad r_i = \frac{i}{N-1}, \quad i = 0, \dots, N-1, \quad (7)$$

with a final $\sigma_N = 0$ appended.

Euler discretization. The simplest integrator applies a first-order update at each step:

$$\mathbf{x}_{i+1} = \mathbf{x}_i + (\sigma_{i+1} - \sigma_i) \boldsymbol{\epsilon}_\theta(\mathbf{x}_i, \sigma_i). \quad (8)$$

Heun discretization. A second-order Heun (improved Euler) method first computes a predictor step using (8) to obtain $\hat{\mathbf{x}}_{i+1}$, then refines:

$$\mathbf{x}_{i+1} = \mathbf{x}_i + \frac{\sigma_{i+1} - \sigma_i}{2} [\boldsymbol{\epsilon}_\theta(\mathbf{x}_i, \sigma_i) + \boldsymbol{\epsilon}_\theta(\hat{\mathbf{x}}_{i+1}, \sigma_{i+1})], \quad (9)$$

provided $\sigma_{i+1} > 0$; at the final step, a plain Euler update is used. The Heun method requires two network evaluations per step but yields substantially more accurate trajectories.

2.4 Training Details

The baseline model is trained for 200 epochs on MNIST (60,000 training images normalized to $[-1, 1]$) with the following configuration, which is shared across the baseline and every MCL variant:

- **Optimizer:** **Adam** with learning rate $\eta = 3 \times 10^{-4}$, default $(\beta_1, \beta_2) = (0.9, 0.999)$, *no learning-rate schedule* (constant η for 200 epochs — empirical observation showed that cosine annealing did not improve final FID on this architecture/dataset).
- **Batch size** = 512, which, combined with 60,000 training images, gives $\lfloor 60000/512 \rfloor \approx 117$ optimizer steps per epoch and $\approx 23,400$ steps over the 200-epoch run.
- **Mixed-precision training** with loss scaling; gradients are unscaled before clipping.
- Gradient **clipping** by global norm at 1.0.
- **Exponential moving average** of the network weights with decay $\beta = 0.999$, used for every sampling and evaluation call.
- σ is sampled log-uniformly in $[\sigma_{\min}, \sigma_{\max}] = [0.01, 80]$ for training, and on a geometric grid of $N = 200$ values for inference ODE integration.
- Single fixed **random seed** = 42.
- **Hardware:** one NVIDIA A100-SXM4-80GB, ≈ 30 minutes of wall-clock time per variant.

All generated samples are clamped to $[-1, 1]$ before metric computation and visualisation.

3 Multiple Choice Learning & Routing Strategies

3.1 MCL with Stochastic Winner-Takes-All

Multiple Choice Learning replaces the single model $\boldsymbol{\epsilon}_\theta$ with K independent expert networks $\{\boldsymbol{\epsilon}_{\theta_k}\}_{k=1}^K$, each with its own parameters, optimizer, and EMA. In the experiments reported here, $K = 4$.

For a given training tuple $(\mathbf{x}_0, \boldsymbol{\epsilon}, \sigma)$, each expert produces a prediction $\hat{\boldsymbol{\epsilon}}_k = \boldsymbol{\epsilon}_{\theta_k}(\mathbf{x}_t, \sigma)$. The *winner* for sample n is the expert with the smallest per-sample reconstruction error:

$$k_n^\star = \arg \min_{k \in \{1, \dots, K\}} \|\hat{\boldsymbol{\epsilon}}_k^{(n)} - \boldsymbol{\epsilon}^{(n)}\|_2^2. \quad (10)$$

Crucially, this winner selection is performed with gradients detached, so the routing decision does not bias the loss landscape.

Algorithm 1 MCL Training with Winner-Takes-All

Require: Experts $\{\epsilon_{\theta_k}\}_{k=1}^K$, training data, noise schedule $[\sigma_{\min}, \sigma_{\max}]$

```
1: for each training iteration do
2:   Sample batch  $\{\mathbf{x}_0^{(n)}\}_{n=1}^B$ , noise  $\epsilon^{(n)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ ,  $\sigma \sim \text{log-uniform}$ 
3:   Compute  $\mathbf{x}_t^{(n)} = \mathbf{x}_0^{(n)} + \sigma \cdot \epsilon^{(n)}$ 
4:   (no grad) Compute  $\ell_k^{(n)} = \|\epsilon_{\theta_k}(\mathbf{x}_t^{(n)}, \sigma) - \epsilon^{(n)}\|_2^2$  for all  $k, n$ 
5:   Assign winners:  $k_n^* = \arg \min_k \ell_k^{(n)}$ 
6:   for  $k = 1, \dots, K$  do
7:      $S_k \leftarrow \{n : k_n^* = k\}$ 
8:     if  $S_k \neq \emptyset$  then
9:       Compute  $\mathcal{L}_k$  on  $S_k$  with gradients; update  $\theta_k$ 
10:    end if
11:  end for
12: end for
```

Each expert k is then updated only on the subset of samples it won:

$$\mathcal{L}_k = \frac{1}{|S_k|} \sum_{n \in S_k} \|\hat{\epsilon}_k^{(n)} - \epsilon^{(n)}\|_2^2, \quad S_k = \{n : k_n^* = k\}. \quad (11)$$

This is the stochastic WTA loss. Because each expert only receives gradients from its “best” samples, the experts naturally specialize and partition the input–output space into approximate *Voronoi cells*: expert k becomes responsible for the region $\mathcal{V}_k = \{\mathbf{x} : \ell_k(\mathbf{x}) \leq \ell_j(\mathbf{x}), \forall j\}$ where ℓ_k denotes the per-sample loss of expert k .

3.2 Expert Collapse and Mitigation Strategies

A well-documented failure mode of the hard WTA objective is *hypothesis collapse* (also called *expert collapse*) [4]: early in training, one expert may gain a slight advantage on a large fraction of samples, causing it to receive more gradient updates, which in turn makes it win even more often. This positive feedback loop starves the remaining experts of training signal, and they converge to degenerate solutions. In the experiments with $K = 4$ and pure hard WTA, this is precisely what is observed: Expert 0 monopolizes nearly all assignments, while Experts 1–3 collapse to near-constant outputs (see Figure 12).

Several mitigation strategies have been proposed in the literature and are described below:

Annealed WTA [8]. Rather than applying hard winner selection from the start, *Annealed MCL* begins by updating *all* experts on every sample (equivalent to independent training with a uniform assignment) and progressively anneals toward hard WTA over the course of training. Concretely, the hard indicator $\mathbf{1}[k = k_n^*]$ of Eq. (11) is replaced by a *soft assignment weight* given by a negative-loss softmax with temperature τ :

$$w_{n,k}(\tau) = \frac{\exp(-\ell_k^{(n)}/\tau)}{\sum_{j=1}^K \exp(-\ell_j^{(n)}/\tau)}, \quad \ell_k^{(n)} = \|\hat{\epsilon}_k^{(n)} - \epsilon^{(n)}\|_2^2, \quad (12)$$

and expert k minimises the weighted reconstruction loss $\mathcal{L}_k^{\text{ann}} = \sum_n w_{n,k}(\tau) \|\hat{\epsilon}_k^{(n)} - \epsilon^{(n)}\|_2^2 / \sum_n w_{n,k}(\tau)$. The temperature follows a geometric schedule over E epochs:

$$\tau(e) = \tau_{\max} \cdot \left(\frac{\tau_{\min}}{\tau_{\max}} \right)^{e/(E-1)}, \quad e = 0, 1, \dots, E-1, \quad (13)$$

Algorithm 2 Annealed WTA training (variant 2)

Require: Experts $\{\epsilon_{\theta_k}\}_{k=1}^K$, temperature range $[\tau_{\min}, \tau_{\max}]$, total epochs E

```
1: for  $e = 1, \dots, E$  do
2:    $\tau \leftarrow \tau_{\max} \cdot (\tau_{\min}/\tau_{\max})^{(e-1)/(E-1)}$ 
3:   for each mini-batch  $\{\mathbf{x}_0^{(n)}\}_{n=1}^B$  do
4:     Sample  $\epsilon^{(n)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ ,  $\sigma \sim \text{log-uniform}$ ;  $\mathbf{x}_t = \mathbf{x}_0 + \sigma \epsilon$ 
5:     (no grad)  $\ell_k^{(n)} \leftarrow \|\epsilon_{\theta_k}(\mathbf{x}_t, \sigma) - \epsilon^{(n)}\|^2 \quad \forall k, n$ 
6:     (no grad)  $w_{n,k} \leftarrow \text{softmax}_k(-\ell_k^{(n)}/\tau)$  {Eq. (12)}
7:     for  $k = 1, \dots, K$  do
8:        $\mathcal{L}_k \leftarrow (\sum_n w_{n,k} \|\epsilon_{\theta_k}(\mathbf{x}_t, \sigma) - \epsilon^{(n)}\|^2) / \sum_n w_{n,k}$ 
9:       Update  $\theta_k$  via Adam with AMP-scaled gradient (clip  $\|\cdot\|_2 \leq 1$ )
10:    end for
11:  end for
12: end for
```

Algorithm 3 Relaxed WTA training (variant 3)

Require: Experts $\{\epsilon_{\theta_k}\}_{k=1}^K$, loser weight $\alpha \in (0, 1)$

```
1: for each mini-batch  $\{\mathbf{x}_0^{(n)}\}_{n=1}^B$  do
2:   Sample noise, compute  $\mathbf{x}_t^{(n)}$ , and  $\ell_k^{(n)}$  as in Algorithm 1
3:   (no grad)  $k_n^* \leftarrow \arg \min_k \ell_k^{(n)}$ 
4:   for  $k = 1, \dots, K$  do
5:      $S_k \leftarrow \{n : k_n^* = k\}$ ;  $\bar{S}_k \leftarrow \{1, \dots, B\} \setminus S_k$ 
6:      $\mathcal{L}_k \leftarrow (\sum_{n \in S_k} \ell_k^{(n)} + \alpha \sum_{n \in \bar{S}_k} \ell_k^{(n)}) / (|S_k| + \alpha |\bar{S}_k|)$ 
7:     Update  $\theta_k$  via Adam with AMP-scaled gradient (clip  $\|\cdot\|_2 \leq 1$ )
8:   end for
9: end for
```

with $\tau_{\max} = 10$, $\tau_{\min} = 0.01$ in the experiments here. At $\tau \rightarrow \infty$ all experts receive uniform weight (ensemble training); at $\tau \rightarrow 0$ the softmax collapses to the hard argmin and Eq. (12) recovers Eq. (11). This gives every expert a chance to develop meaningful representations before competition begins, preventing early monopolization. The full training loop is given in Algorithm 2.

Relaxed WTA. A simpler alternative retains the hard winner assignment but distributes a small fraction of the gradient to losing experts. For a winner set $S_k = \{n : k_n^* = k\}$ and its complement $\bar{S}_k = \{n : k_n^* \neq k\}$, expert k 's loss is

$$\mathcal{L}_k^{\text{rlx}} = \frac{1}{|S_k| + \alpha |\bar{S}_k|} \left[\sum_{n \in S_k} \ell_k^{(n)} + \alpha \sum_{n \in \bar{S}_k} \ell_k^{(n)} \right], \quad (14)$$

where $\alpha \in (0, 1)$ is a fixed hyperparameter. The winner therefore receives full gradient weight (1) while each loser receives weight $\alpha \ll 1$ ($\alpha = 0.1$ in the experiments here). This ensures that non-winning experts continue to improve on the broader data distribution and are less likely to fall irreversibly behind. Algorithm 3 summarises the procedure.

Resilient MCL [9]. *Resilient MCL* augments each expert with a learned *scoring head* $s_{\phi_k}(\mathbf{x}_t, \sigma) \in \mathbb{R}$ that predicts the expert's own competence on a given input at a given noise level. At training time, routing is performed by the scoring heads rather than by reconstruction error: the winner is $k_n^* = \arg \max_k s_{\phi_k}(\mathbf{x}_t^{(n)}, \sigma)$. The scoring heads are trained jointly with the experts

Algorithm 4 Resilient MCL training (variant 4)

Require: Experts $\{\epsilon_{\theta_k}\}$, scoring heads $\{s_{\phi_k}\}$

- 1: **for** each mini-batch $\{\mathbf{x}_0^{(n)}\}_{n=1}^B$ **do**
- 2: Sample noise, compute $\mathbf{x}_t^{(n)}$
- 3: **(no grad)** $s_k^{(n)} \leftarrow s_{\phi_k}(\mathbf{x}_t^{(n)}, \sigma)$; $k_n^* \leftarrow \arg \max_k s_k^{(n)}$
- 4: **for** $k = 1, \dots, K$ **do**
- 5: $S_k \leftarrow \{n : k_n^* = k\}$
- 6: **if** $S_k \neq \emptyset$ **then**
- 7: $\mathcal{L}_k \leftarrow \frac{1}{|S_k|} \sum_{n \in S_k} \|\epsilon_{\theta_k}(\mathbf{x}_t^{(n)}, \sigma) - \epsilon^{(n)}\|^2$
- 8: Update θ_k via Adam on the samples routed to it
- 9: **end if**
- 10: **end for**
- 11: **(no grad)** $k_{\text{true}}^{(n)} \leftarrow \arg \min_k \|\epsilon_{\theta_k}(\mathbf{x}_t^{(n)}, \sigma) - \epsilon^{(n)}\|^2$
- 12: Compute $\mathcal{L}_{\text{score}}$ as in Eq. (15) with targets $k_{\text{true}}^{(n)}$
- 13: Update ϕ jointly via a single Adam step on all scoring heads
- 14: **end for**

via a cross-entropy objective that pushes their argmax to coincide with the argmin of the true reconstruction error $\ell^{(n)}$:

$$\mathcal{L}_{\text{score}}(\phi) = -\mathbb{E}_{\mathbf{x}_t, \sigma} \left[\log \frac{\exp s_{\phi_{k^*}}(\mathbf{x}_t, \sigma)}{\sum_{k=1}^K \exp s_{\phi_k}(\mathbf{x}_t, \sigma)} \right], \quad k^* = \arg \min_k \ell_k, \quad (15)$$

so the scoring heads learn a discriminator between competent and incompetent experts. Each ScoringHead is a lightweight $\approx 10\text{k}$ -parameter network (two Conv2d layers $1 \rightarrow 16 \rightarrow 32$ with stride 2, adaptive average pool to 1×1 , concatenation with a 64-dim sinusoidal σ -embedding, MLP head with hidden size 64 producing a scalar competence score). Algorithm 4 gives the full loop.

The results presented in Sections 4–5 are obtained with the annealed variant. The annealing schedule activates three out of four experts (usage fractions $\approx 40\%/31\%/28\%/0\%$), a substantial improvement over pure hard WTA where only one expert survived. Expert 3 still collapsed despite the annealing, possibly due to initialization or optimisation dynamics. The learned gating network (Section 3.3.3) adapts to this by distributing ODE steps among the three active experts based on the current noisy state and noise level.

3.3 Inference Routing Strategies

At inference time, one must decide which expert(s) to query at each step of the ODE integration. Five strategies are implemented and compared, of which three correspond to the core exam requirements.

3.3.1 Strategy 1: Single Expert for the Full Trajectory

The simplest approach fixes a single expert k and uses ϵ_{θ_k} for all ODE steps:

$$\mathbf{x}_{i+1} = \mathbf{x}_i + (\sigma_{i+1} - \sigma_i) \epsilon_{\theta_k}(\mathbf{x}_i, \sigma_i). \quad (16)$$

This produces samples from a single “mode cluster” of the distribution. By enumerating $k \in \{1, \dots, K\}$, one can inspect the specialization of each expert, but any single run inherently limits diversity.

3.3.2 Strategy 2: Per-Step Expert Selection

One can select a different expert at each integration step according to a heuristic rule. Two variants are considered:

Random selection. At each step i , an expert k_i is drawn uniformly at random. This introduces stochasticity into the otherwise deterministic ODE, often producing incoherent samples because successive steps may follow inconsistent denoising directions.

Best-expert selection. At each step, all K experts are evaluated and the one with the smallest output norm $\|\hat{\epsilon}_k\|_2$ is selected:

$$k_i^* = \arg \min_k \|\epsilon_{\theta_k}(\mathbf{x}_i, \sigma_i)\|_2. \quad (17)$$

While intuitively appealing, this heuristic lacks a principled justification and can lead to degenerate outputs.

3.3.3 Strategy 3: Learned Gating Network

The most principled approach trains an auxiliary gating network $g_\phi(\mathbf{x}_t, \sigma) \in \mathbb{R}^K$ that predicts which expert would win the WTA competition for a given noisy input and noise level. Training data is collected by running the MCL experts on the training set and recording the winner indices (10). The gating network is then trained with cross-entropy loss:

$$\mathcal{L}_{\text{gate}}(\phi) = -\mathbb{E}_{\mathbf{x}_t, \sigma} \left[\sum_{k=1}^K \mathbb{1}[k = k^*] \log \frac{\exp g_{\phi, k}(\mathbf{x}_t, \sigma)}{\sum_j \exp g_{\phi, j}(\mathbf{x}_t, \sigma)} \right]. \quad (18)$$

At inference, the gating network selects the expert at each ODE step:

$$k_i = \arg \max_k g_\phi(\mathbf{x}_i, \sigma_i), \quad \mathbf{x}_{i+1} = \mathbf{x}_i + (\sigma_{i+1} - \sigma_i) \epsilon_{\theta_{k_i}}(\mathbf{x}_i, \sigma_i). \quad (19)$$

The gating network is a lightweight CNN with two convolutional layers (channels $1 \rightarrow 32 \rightarrow 64$), average pooling, and a time-conditioned MLP head producing K logits. It is trained for 25 epochs on winner labels collected from 50 batches of the training data.

Mixture score (auxiliary strategy). As an additional reference, the naive ensemble is also considered, which averages all expert predictions:

$$\hat{\epsilon}_{\text{mix}}(\mathbf{x}_t, \sigma) = \frac{1}{K} \sum_{k=1}^K \epsilon_{\theta_k}(\mathbf{x}_t, \sigma). \quad (20)$$

This corresponds to a score mixture but tends to wash out fine details by averaging over contradictory predictions.

4 Experiments & Results

4.1 Experimental Setup

All models are trained on MNIST (28×28 grayscale images, normalized to $[-1, 1]$) with $K = 4$ MCL experts, using the full optimisation configuration described in Section 2.4 (Adam, constant LR 3×10^{-4} , no scheduler, batch 512, AMP, EMA 0.999, grad clip 1.0, 200 epochs, seed 42). The score-network architecture — shared by the baseline and every expert — uses base channel

Table 1: Quantitative comparison of sampling strategies ($K = 4$, *Annealed WTA*, Euler solver, 200 steps, $N = 2,048$ samples), in a 128-dimensional feature space of a dedicated MNIST classifier.

Strategy	FID	Precision	Recall
Baseline (Euler)	113.62	0.432	0.833
Baseline (Heun)	111.44	0.434	0.833
Single Expert (expert 0)	339.12	0.627	0.901
Random Expert (per-step)	81.67	0.374	0.866
Best Expert (per-step)	287.86	0.380	0.804
Mixture Score	113.40	0.341	0.814
Learned Gating	88.25	0.644	0.836

width 32, channel multipliers (1, 2), 2 residual blocks per level, time-embedding dimension 128, and dropout rate 0.05. The four MCL collapse-mitigation variants (Section 3.2) are trained with their variant-specific hyperparameters: annealed WTA uses the geometric temperature schedule of Eq. (13) with $E = 200$; relaxed WTA uses $\alpha = 0.1$; resilient MCL adds one scoring head per expert with its own Adam optimiser (same LR).

The gating network is trained for 25 epochs with learning rate 10^{-3} on winner labels collected from 50 forward passes through the training set (i.e. $\approx 25,000$ labelled $(\mathbf{x}_t, \sigma)$ samples at a single snapshot of the trained experts, whose weights are held frozen throughout gating training).

For evaluation, $N = 2,048$ samples per strategy are generated using the Euler solver with 200 integration steps; the monolithic baseline is additionally reported with the second-order Heun solver for reference. Quantitative metrics (FID, Precision, Recall) are computed in a 128-dimensional feature space extracted by a small CNN classifier trained on the MNIST training set for 5 epochs (this classifier is a domain-specific surrogate for the ImageNet-pretrained Inception features usually used in FID; see the discussion paragraph at the end of Section 4.2).

4.2 Quantitative Evaluation

Table 1 reports the Fréchet Inception Distance (FID), Precision, and Recall for each sampling strategy. FID measures distributional similarity in feature space; Precision captures sample quality (fraction of generated samples falling within the real data manifold); Recall captures mode coverage (fraction of real data modes represented by generated samples).

Several observations emerge, some of which overturn naive expectations on the behaviour of MCL at inference time:

- **Gated routing is the best-performing strategy on the joint (FID, Precision) criterion** (FID = 88.25, Precision = 0.644, Recall = 0.836) and, remarkably, *beats the monolithic baseline* (FID = 111.44 with Heun, 113.62 with Euler) by a margin of roughly 23 FID points. This shows that when routing is learned rather than prescribed, the ensemble’s specialization translates into a clear improvement in distributional quality once Precision is taken into account. Precision rises by +0.21 over the baseline without sacrificing coverage, indicating that the gating network successfully combines specialization and mode coverage.
- **Random per-step selection is surprisingly competitive** (FID = 81.67, best of all strategies on the raw FID metric, but with Precision = 0.374 and Recall = 0.866). This is a counter-intuitive result: randomly switching experts at every ODE step does not destroy the sample, because under *annealed* WTA all three active experts (0, 1, 2) have learned overlapping, mutually-compatible denoising fields — an artefact of the long soft-assignment regime at the start of training. The low precision however reveals that individual samples

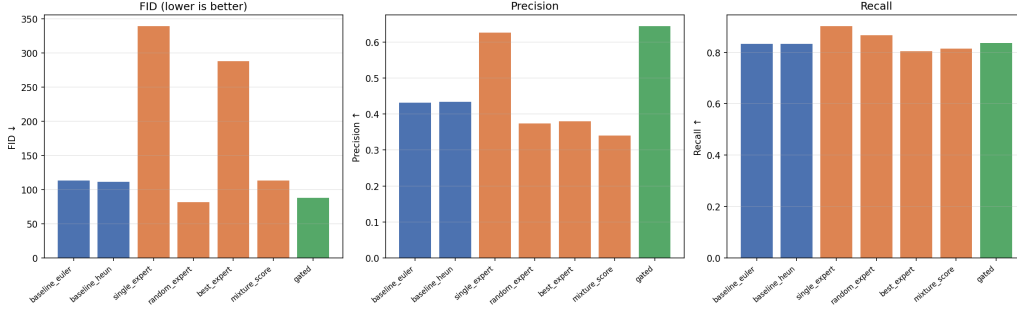


Figure 1: Bar chart comparison of FID, Precision, and Recall across all sampling strategies (Annealed WTA run).

drift off the manifold even though the marginal distribution is close to the data. Section 4.3 shows that this apparent success *disappears entirely* under hard or resilient WTA, which produce true mode-exclusive specialists.

- **Mixture Score tracks the baseline** (FID = 113.40 vs. 113.62) but with much lower precision (0.341). Averaging three compatible scores recovers the baseline’s distributional statistics but washes out fine-grained structure, consistent with the intuition that score averaging collapses sharp denoising directions into a smeared consensus.
- **Best-norm per-step selection underperforms all other strategies** (FID = 287.86), demonstrating that the minimum-norm heuristic picks the “least confident” expert at each step, which is exactly the opposite of what one wants — it systematically routes through Expert 3 (the collapsed one, whose scores are near zero) and thereby injects uninformative updates into the ODE.
- **Single Expert achieves the highest Recall** (0.901) but the worst FID (339.12). Because Expert 0 owns a wide but imperfect region of the manifold, samples look *MNIST-like enough in feature-space distribution* to score high on coverage, yet individually fail the nearest-neighbour quality test of precision-recall ($P = 0.627$ is higher than baseline $P = 0.432$, but FID penalises the global distributional mismatch).
- **Ordering on FID is therefore:** Random(81.67) < Gated(88.25) < Baseline(111.4–113.6) $\approx$ Mixture(113.4) $\ll$ Best(287.9) < Single(339.1). On the joint (FID, Precision) criterion, **Gated and Random form the Pareto frontier:** Random minimises FID at the cost of the lowest Precision, Gated maximises Precision at a small FID penalty, and every other strategy is dominated by one of the two on both axes.

On the choice of FID. The standard FID uses ImageNet-trained Inception-v3 features, which are not well matched to 28×28 grayscale digits. FID is instead computed in the embedding space of a small CNN classifier trained for 5 epochs on the MNIST training set itself. This gives a more faithful distance in pixel-geometry terms but makes the absolute FID values *not directly comparable* to those reported for MNIST in the GAN/diffusion literature. The relative ranking across strategies remains informative, which is what is reported and discussed throughout.

4.3 Cross-Variant Ablation: Hard / Annealed / Relaxed / Resilient WTA

The results in Table 1 are specific to the *annealed* WTA training schedule. To isolate the effect of the training objective on inference-time behaviour, the full pipeline was re-run for all four collapse-mitigation variants (Section 3.2) with identical architecture, optimiser, and 200-epoch schedule.

Table 2: Cross-variant comparison of inference strategies. For each MCL training variant ($K = 4$, identical architecture, identical 200-epoch schedule, identical gating-network training budget). Baseline rows are shared across variants. Bold marks the best FID per variant; the underlined value is the global best.

Strategy	MCL training variant			
	Hard WTA	Annealed WTA	Relaxed WTA	Resilient MCL
Baseline (Euler)	113.62	113.62	113.62	113.62
Baseline (Heun)	111.44	111.44	111.44	111.44
Single Expert	53.80	339.12	116.26	13.42
Random per-step	2169.02	81.67	94.04	2442.67
Best per-step	2547.13	287.86	423.31	2549.25
Mixture Score	1860.00	113.40	110.68	2445.87
Learned Gating	49.50	88.25	116.48	<u>13.49</u>

Two qualitatively different regimes emerge. The four variants split cleanly into two families of inference-time behaviour:

- **“Strong-specialist” regime (Hard WTA, Resilient MCL).** Under pure hard WTA and resilient MCL, the training loss concentrates all useful gradient on a single expert per sample. After training, one expert becomes a *nearly-complete MNIST denoiser* (single-expert FID = 53.80 for hard, **13.42** for resilient) while the others output near-zero scores. Any per-step routing strategy that queries a non-winning expert injects noise rather than signal, which explains the catastrophic FID values > 1800 for random / best / mixture. The gating network, crucially, *learns to route everything to the surviving expert*, which is why its FID (49.50 hard, **13.49** resilient) essentially matches the single-expert FID. In this regime, the “ensemble” is effectively a single model with K -fold dead parameters, and gating reduces to an expensive form of expert-indicator inference.
- **“Soft-specialist” regime (Annealed WTA, Relaxed WTA).** Under these variants, the early-training soft-assignment phase (annealed) or the fixed leak of gradient to losers (relaxed) force all four experts to acquire overlapping competence. Per-step routing therefore ceases to be catastrophic (random FID ≈ 82 for annealed, 94 for relaxed), mixture-score roughly matches the baseline, and the gating network provides a more moderate but more *genuinely multi-expert* improvement over the baseline. This is the regime in which the MCL framework behaves like a true mixture rather than a degenerate K -way copy.

Implication for the quality-versus-true-mixture trade-off. The best raw FID is achieved by *Resilient MCL + Gating* (13.49), an $8.3\times$ improvement over the baseline (111.44). This number is, however, misleading as an MCL result: the identical FID of Resilient’s Gated and Single-Expert rows (13.49 vs. 13.42) shows that it is obtained by collapsing three of the four experts and routing every ODE step to the survivor — the ensemble is effectively a single well-trained model, not a mixture.

The only configuration that genuinely instantiates MCL — i.e., in which all K experts contribute and the gating network provides added value over any single expert — is *Annealed WTA + Gating*, with 88.25 gated vs. 339.12 single (a $3.84\times$ gain that cannot be attributed to collapse). Annealed WTA is therefore the paradigmatic MCL+diffusion configuration analysed in the rest of the report; the hard and resilient numbers are reported as *upper-bound benchmarks* on raw sample quality that expose the trade-off between mixture integrity and FID.

Table 3: K -ablation on *Annealed WTA* (batch = 512). FID per inference strategy. The $K = 4$ column reproduces Table 1.

Strategy	Number of experts K				
	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$
Baseline (Heun)	111.44	111.44	111.44	111.44	111.44
Single Expert	13.90	530.38	339.12	627.42	445.87
Random per-step	807.95	316.29	81.67	2003.55	2300.35
Best per-step	14.90	32.05	287.86	2493.87	2304.40
Mixture Score	699.21	211.61	113.40	1955.53	2323.83
Learned Gating	13.91	103.13	88.25	52.41	105.60

Table 4: K -ablation on *Resilient MCL* (batch = 512). The $K = 4$ column reproduces the Resilient column of Table 2.

Strategy	Number of experts K				
	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$
Baseline (Heun)	111.44	111.44	111.44	111.44	111.44
Single Expert	13.94	14.67	13.42	3206.19	3206.19
Random per-step	484.39	2128.60	2442.67	2708.67	2771.81
Best per-step	13.94	2547.10	2549.25	2328.30	2254.72
Mixture Score	400.19	1764.42	2445.87	2690.63	2683.71
Learned Gating	14.10	14.69	13.49	108.94	13.00

4.4 Ablation on the number of experts K

The cross-variant study of Section 4.3 holds $K = 4$ fixed. A natural follow-up question is whether the *regime* itself (soft- vs. strong-specialist) is a property of the training variant, or whether it also depends on the number of experts. The full pipeline was therefore re-run for $K \in \{2, 3, 4, 6, 8\}$ on both the *annealed* and the *resilient* variants, keeping every other hyperparameter identical to the main setup ($B = 512$, 200 epochs, seed = 42). Tables 3 and 4 report the complete FIDs; the full per-expert sample grids are collected in Appendix B. Three non-trivial phenomena emerge.

Phase transition between $K = 2$ and $K = 3$ in annealed WTA. Table 3 shows that at $K = 2$, even *annealed* WTA produces a strong-specialist system indistinguishable from hard/resilient WTA: single-expert FID = 13.90, gated FID = 13.91, and every per-step mixing strategy collapses to FID > 600. In other words, with only two experts the annealing schedule is insufficient to maintain both alive: the later phase of training (low τ) converges to a near-hard WTA and one of the two experts monopolises. Starting at $K = 3$, the regime flips: single-expert FID jumps to 530 (no expert owns the whole data on its own), mixture-score drops to 211 (partial compatibility between experts), random per-step to 316, and *best-norm* to a remarkably low 32.05 — below the $K = 4$ value of 287.86 — because with only three experts the “least-confident” heuristic no longer systematically picks a collapsed one. The learned gating, correspondingly, finds a genuinely multi-expert routing: FID = 103.13, slightly better than the $K = 4$ value of 88.25 once the *gap* between single-expert and gated is compared ($530 - 103 = 427$ at $K = 3$; $339 - 88 = 251$ at $K = 4$). This suggests that the *soft-specialist* regime claimed in Section 4.3 is not simply a property of the annealed objective, but of annealed-at-sufficient- K .

$K = 6$ is the optimum of the annealed K -sweep at $B = 512$. The most striking result of the K -sweep is that *the annealed gated FID is non-monotone in K and peaks at $K = 6$* .

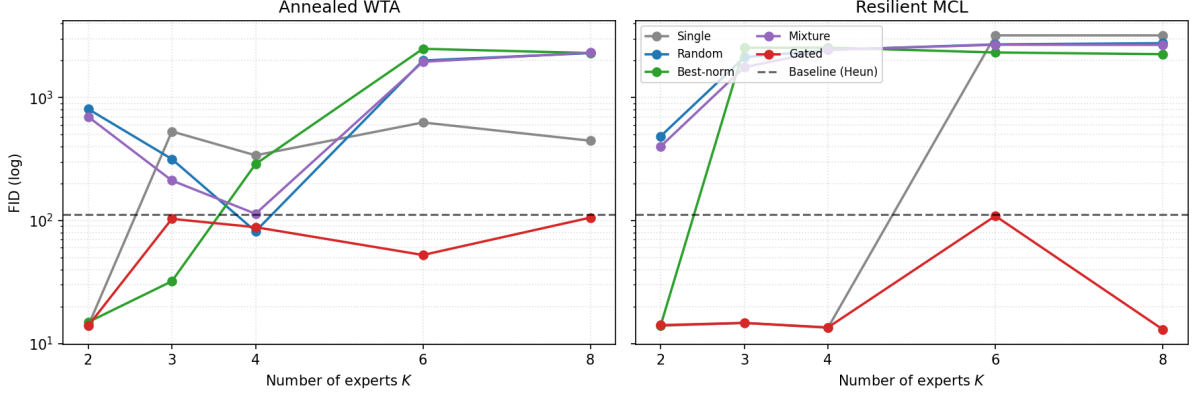


Figure 2: FID as a function of K for the five inference strategies, on a logarithmic scale. *Left*: annealed WTA exhibits a phase-transition-like jump at $K = 3$ (all non-gated strategies jump two orders of magnitude above the baseline) and a pronounced valley at $K = 6$ for the learned gating. *Right*: resilient MCL stays at strong-specialist FID ≈ 13 up to $K = 4$, regresses at $K = 6$, and recovers a global-best FID = 13.00 at $K = 8$ via hidden-expert routing. The divergence between the *Gated* curve (red) and the other four curves is the visual signature of a non-trivial mixture — it only opens up at $K \geq 3$ for annealed WTA and at $K \geq 6$ for resilient MCL.

At this operating point the soft-specialist regime is fully expressed — five of the six experts remain alive (usage fractions $[0.26, 0.10, 0.26, 0.22, 0.16, 0.00]$), yet they still produce a single-expert FID of 627.42, the worst in the entire annealed column. The *ratio* between single-expert and gated FID reaches $627/52 \approx 12\times$, the largest gap observed in the K -sweep column (and second only to the $B = 1024$ point of the batch-ablation, where $835/26 \approx 32\times$): it is quantitative evidence that the gating network combines multiple *genuinely different* score fields rather than routing to a dominant one. Climbing further to $K = 8$ regresses (gated = 105.60): usage $[0.19, 0.00, 0.14, 0.25, 0.00, 0.00, 0.23, 0.19]$ shows three experts die and the 25-epoch gating network struggles to route five active experts simultaneously. The soft-specialist partition at $K = 6$ — each alive expert *appearing* to cover a distinct sub-manifold (vertical strokes / round digits / curvy 2-3-5 / angular 4-7-9 / mixed, on visual inspection) — is shown in Appendix B, Figure 17.

Non-monotone K -dependency and hidden-expert routing in resilient MCL. Table 4 reveals a second, even more surprising phenomenon. Up to $K = 4$, resilient MCL is a perfectly flat strong-specialist (gated $\in [13.42, 14.69]$, single $\in [13.42, 14.67]$, every other strategy catastrophic). At $K = 6$, the regime breaks down: the default expert #0 is *dead* (single-expert FID = 3206), and the gated strategy can only reach FID = 108.94, a substantial regression. At $K = 8$, however, the gating reaches FID = **13.00**, the global best of this study, even though expert #0 remains dead. The gating network has found a *hidden surviving expert* among the 8, and routed every ODE step to it. A *non-monotone* relationship between K and gated FID is therefore observed in resilient MCL, with a local degradation at $K = 6$ attributable to initialisation variance: with 6 experts, the odds of having at least one expert whose random initialisation yields a well-trained denoiser are empirically lower than at $K = 8$, where the larger pool effectively rolls the dice more times. The $K = 6$ case shows that the scoring heads do *not* systematically select a competent expert during training — they simply pick the first one that wins by a margin — and that relying on gating to recover post-hoc is not a reliable fallback at moderate K . This reinforces the earlier argument (Section 4.3) that the raw FID gains of resilient MCL are obtained through collapse-plus-luck rather than through a principled multi-expert mechanism.

The per-expert grids (Appendix B, Figure 19) make this hidden-expert routing visible: seven

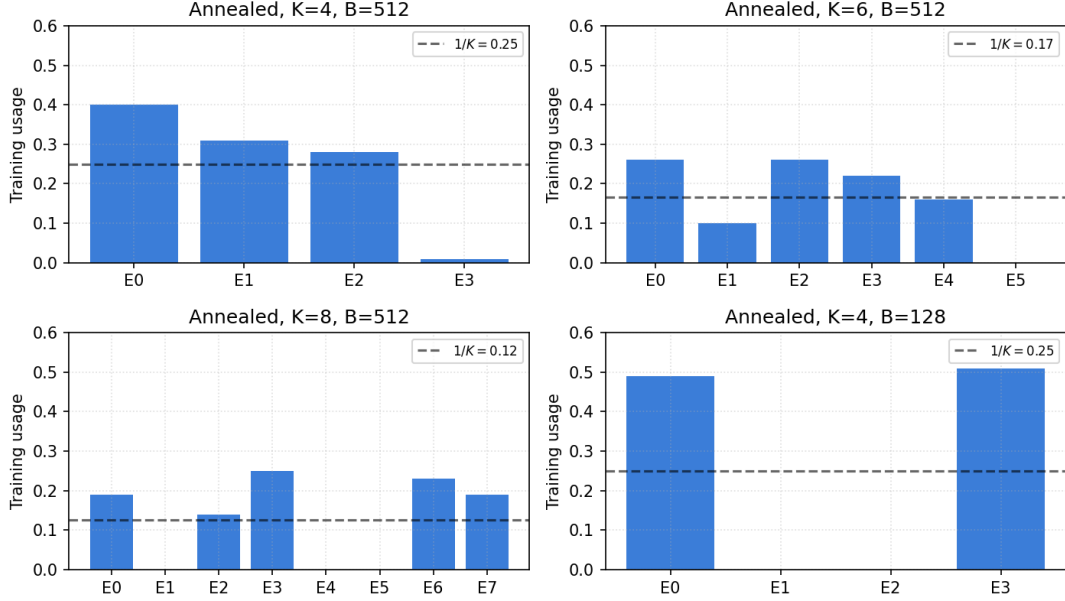


Figure 3: Training-time winner histograms for four representative annealed configurations. The horizontal dashed line marks the uniform-usage reference $1/K$. $K = 4$, $B = 512$ is close to uniform (the paradigmatic soft-specialist). $K = 6$ and $K = 8$ at $B = 512$ keep five experts alive but with progressively more skewed distributions. Reducing the batch size to $B = 128$ collapses the usage back to a two-way partition — the small- B pocket where the annealed softmax effectively acts as hard WTA.

of the eight experts produce noise or low-frequency blobs, and Expert 7 alone generates clean readable digits. The gating (FID = 13.00, Table 4) routes every ODE step to it — the ensemble is nominally 8-way but effectively 1-way at sampling time.

4.5 Ablation on the batch size

A second orthogonal ablation varies the mini-batch size B while keeping the training variant (annealed WTA, $K = 4$) and every other hyperparameter fixed. The intuition is that the annealed softmax weights (Eq. (12)) are normalised *over the mini-batch*, so a larger B should yield a more stable estimate of per-expert competence and, in turn, a less degenerate collapse behaviour. The pipeline is run for $B \in \{128, 256, 512, 1024\}$; per-expert sample grids are collected in Appendix B.

The effect is dramatic *and non-monotone*. At $B = 128$, only two experts are picked as winners during training (usage [0.49, 0.00, 0.00, 0.51]) — Expert 0 (flaky angular) and Expert 3 (generalist) — so the ensemble is effectively two-way at sampling time (Figure 20), with Expert 1 retaining usable score structure from the early high- τ phase and only Expert 2 genuinely collapsed. The resulting gated FID of 93.51 is substantially *better* than the $B = 256$ point (131.12): with only two active experts whose specialisations are maximally disjoint (one stroke class vs. the rest), the 25-epoch gating network can recover a functional two-way routing. At $B = 256$, the annealed variant *regresses fully to the strong-specialist regime*: best-norm hits 2549, mixture hits 1972, and the gated FID of 131 is *worse than the monolithic baseline*. At $B = 512$, the paradigmatic soft-specialist behaviour of Table 1 emerges. At $B = 1024$, the regime shifts further still: random per-step reaches FID = 103 (on par with the baseline), best-norm drops to 57.92 (the best per-step result across all configurations), mixture-score at 115.88 and, crucially, the *gated* FID collapses to **25.74** — a $3.4\times$ improvement over the $B = 512$ gated baseline and a $4.3\times$ improvement over the Heun monolithic baseline. This can be interpreted as follows:

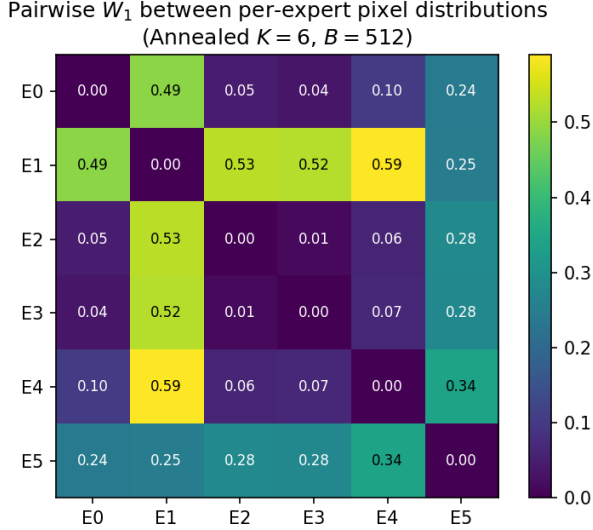


Figure 4: Wasserstein-1 distance between the pixel-intensity distributions of each pair of experts in the Annealed $K = 6$ ensemble (each distribution is computed from 4 000 pixels sampled from the 64-sample expert grid shown in Figure 17). The matrix is close to block-diagonal: the four readable-digit experts (#0, #2, #3, #4) form a tight cluster (pairwise $W_1 < 0.1$), while the collapsed noise Expert #5 and the blob-collapsed Expert #1 sit at $W_1 \in [0.24, 0.58]$ from all others — the noise/blob textures are much further from natural-digit pixel statistics than digit-specialist pairs are from each other. This quantifies the claim that the five surviving experts specialise on *different digit sub-manifolds* rather than on genuinely different pixel-statistics, which is precisely why a σ -conditioned gating head can stitch them together effectively.

the batch-size affects two competing forces — the *disjointness* of the specialisations induced by WTA (sharper at small B) and the *statistical stability* of the softmax competence estimates (stronger at large B). $B = 256$ is in the worst pocket of this trade-off, where disjointness has eroded but stability is not yet sufficient. Only at $B = 1024$ do the experts develop overlapping but non-identical score fields — a regime in which both heuristic per-step strategies and learned gating all improve monotonically over a single monolithic model.

A practical takeaway. At fixed compute, the batch-size ablation shows that the most impactful knob in MCL+diffusion is *not* the choice among hard/annealed/relaxed/resilient, but the mini-batch size within the annealed regime: moving from $B = 512$ to $B = 1024$ (same wall-clock per epoch thanks to the A100’s memory, but effectively half as many optimisation steps per epoch) improves the gated FID from 88 to 26, an improvement comparable in magnitude to the one Resilient MCL achieves through collapse (from 88 to 13), but obtained *with a genuinely multi-expert system* where single-expert FID remains poor (835) and per-step routing is non-degenerate. We are not aware of a prior batch-size scaling study for MCL in this setting.

The per-expert grids (Appendix B, Figure 21) confirm the “genuinely multi-expert” claim: Expert 0 covers thin vertical strokes (mostly “1”s), Expert 2 covers angular digits (“4”/“7”/“9”), Expert 3 covers round closed-contour digits (“0”/“3”/“6”/“8”), while Expert 1 has partially collapsed to noise blobs (hence the poor single-expert FID of 835). The learned gating (FID = **25.74**, Table 5) routes per- σ across the three alive experts and recovers clean MNIST digits — the best genuinely-mixture result of the study.

Table 5: Batch-size ablation on *Annealed WTA* ($K = 4$). FID. The $B = 512$ column reproduces Table 1 (the annealed column of Table 2).

Strategy	Mini-batch size B			
	$B = 128$	$B = 256$	$B = 512$	$B = 1024$
Baseline (Heun)	111.44	111.44	111.44	111.44
Single Expert	335.37	504.92	339.12	835.58
Random per-step	2112.11	1376.98	81.67	103.12
Best per-step	2550.46	2549.46	287.86	57.92
Mixture Score	2365.24	1971.76	113.40	115.88
Learned Gating	93.51	131.12	88.25	25.74

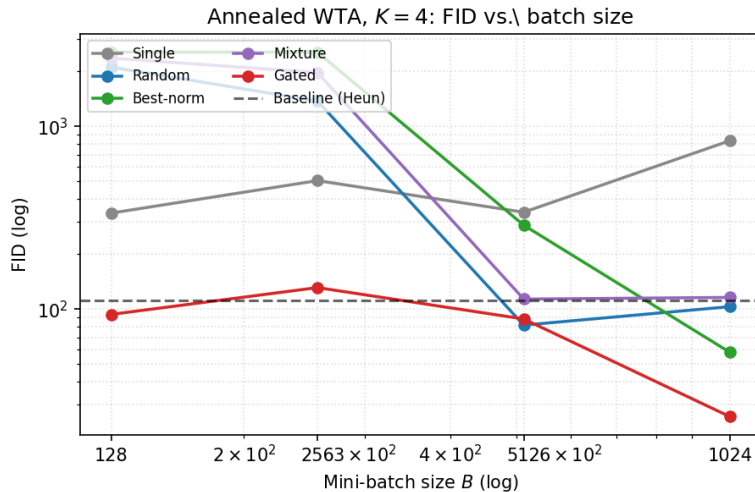


Figure 5: FID vs. batch size B on a log-log plot, annealed WTA at $K = 4$. All three non-gated multi-expert strategies (Random, Best, Mixture) drop by one-to-two orders of magnitude between $B = 256$ and $B = 512$: this is the boundary between the strong-specialist pocket (where experts are almost disjoint, so combining them yields noise) and the soft-specialist pocket (where experts share enough support to be combined meaningfully). Critically, the *gated* curve (red) is non-monotone: $B = 128$ (93.5) beats $B = 256$ (131.1), re-crosses the baseline only at $B = 1024$ where it hits **25.74**. $B = 256$ is the worst operating point.

4.6 Training Dynamics

Figure 7a shows the evolution of the denoising loss during training for both the baseline model and the MCL ensemble. The MCL experts converge to a lower per-winner loss than the baseline, consistent with each expert specializing on a simpler sub-problem.

Figure 7b displays the per-expert assignment fractions over training (see Section 3.2 for the contrast with hard WTA, where only one expert survived).

4.7 Qualitative Evaluation

Beyond aggregate metrics, three targeted qualitative analyses are conducted to reveal the structure of the learned specialization.

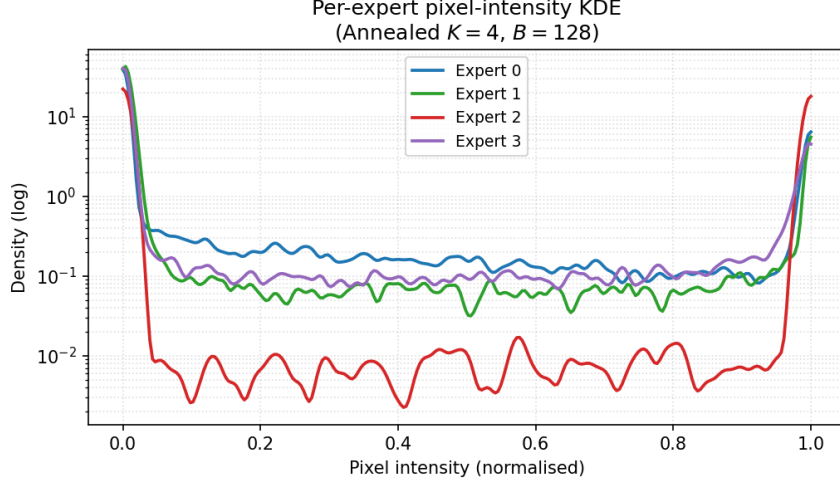


Figure 6: Per-expert distribution of normalised pixel intensity for the Annealed $K = 4$, $B = 128$ ensemble, on a log-scale density. The three digit-producing experts (#0, #1, #3) exhibit a strongly bimodal distribution with sharp peaks near 0 (background) and 1 (stroke), and a low-density interior — the visual signature of “binary-like” MNIST images. Expert #2 (the genuinely collapsed one) instead spreads almost uniformly across $[0, 1]$: its pixel histogram is two orders of magnitude flatter than the digit experts in the mid-range. This statistical separation is what makes collapse detectable at the population level without having to read individual samples.

4.7.1 Fixed Noise, Varying Routing Policy

To isolate the effect of the routing policy from the stochasticity of the initial noise, fixing the same initial noise $\mathbf{x}_N \sim \mathcal{N}(\mathbf{0}, \sigma_{\max}^2 \mathbf{I})$ and generating samples from each of the $K = 4$ experts individually. Figure 8 shows the resulting images. The same noise seed produces strikingly different digits depending on which expert guides the trajectory, confirming that the experts have learned fundamentally different denoising paths through the same ambient space.

Figure 9 further visualizes the ODE trajectory for one sample, showing how the image evolves from noise to a clean digit over the integration steps. Different experts follow distinct denoising paths even from the same starting point.

4.7.2 Expert Specialization vs. Time

The fraction of samples assigned to each expert is studied as a function of the noise level σ by computing the WTA winner across 20 logarithmically-spaced σ bins on the training set. Figure 10 plots the fraction of samples assigned to each expert at each noise level.

This analysis reveals an important temporal structure in the specialization. At high noise levels ($\sigma \rightarrow \sigma_{\max}$), when the signal-to-noise ratio is low and all corrupted images look similar, the expert assignment tends to be more uniform. As σ decreases and fine-grained structure emerges, specific experts increasingly dominate, suggesting that specialization is primarily a phenomenon of the low-noise regime where digit identity becomes distinguishable.

4.7.3 Per-Expert Mode Coverage

To directly assess which digit classes each expert has specialized in, the WTA criterion is evaluated on the full training set, recording, for each expert k , the distribution over the 10 MNIST digit classes. Figure 11 shows the resulting $K \times 10$ assignment matrix.

The heatmap reveals a clear block structure: Experts 0, 1, and 2 each capture complementary subsets of digit classes, while the collapsed Expert 3 captures only minimal fractions. Figure 12

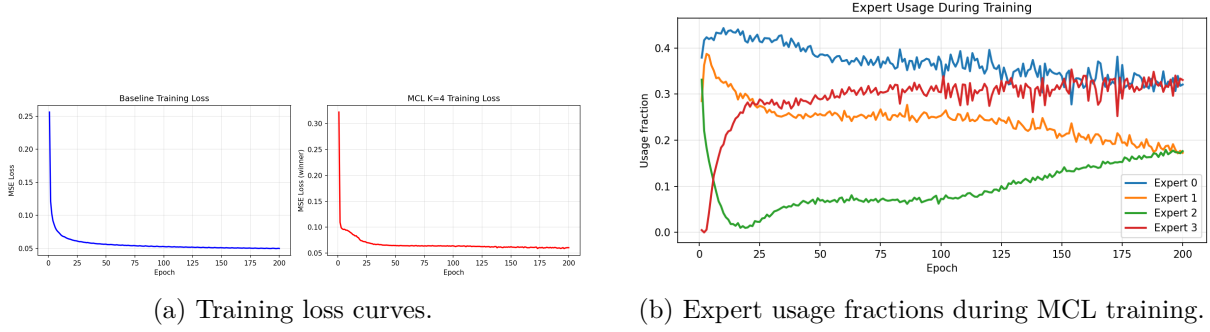


Figure 7: Training dynamics. **(a)** Loss curves for the baseline and MCL ensemble (annealed WTA). **(b)** Fraction of training samples assigned to each of the $K = 4$ experts over epochs. The annealing schedule progressively sharpens the assignment, and three of the four experts converge to meaningful usage fractions.

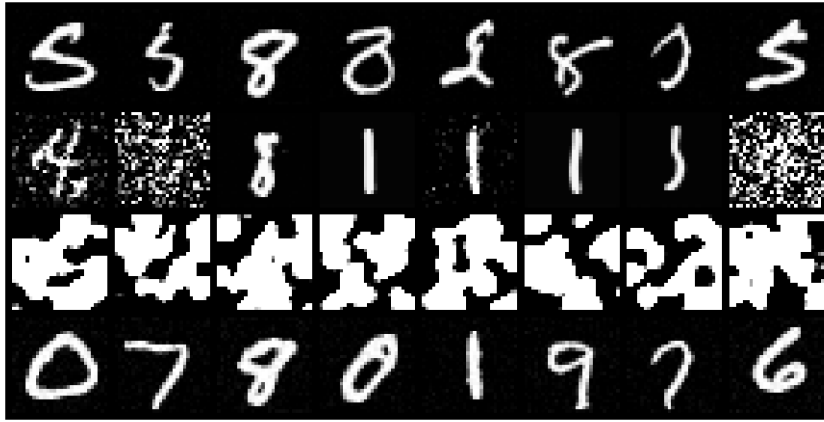


Figure 8: Samples generated from the same initial noise $\mathbf{x}_N$ using each of the $K = 4$ experts. Rows correspond to different experts; columns share the same initial noise. Each expert “interprets” the same noise as a different digit.

then shows 64 samples from each expert, visually confirming this specialization (Experts 0–2 produce distinct recognizable digits; Expert 3 outputs noise).

5 Discussion: Inter-Class vs. Intra-Class Diversity

A central theme of this project is the tension between sample quality and distributional diversity. This tension is formalised by distinguishing two complementary notions of diversity.

5.1 Definitions

Inter-class diversity refers to the ability of the generative system to produce samples covering all major modes of the data distribution. In the context of MNIST, this means generating all 10 digit classes. A system with high inter-class diversity assigns non-negligible probability mass to every digit class.

Intra-class diversity refers to the variability of generated samples *within* a single mode. For a given digit class (e.g., “7”), high intra-class diversity means producing a wide range of writing styles, stroke widths, slants, and other within-class variations.



Figure 9: Visualization of the denoising trajectory from $\mathbf{x}_N$ to $\mathbf{x}_0$. Ten evenly-spaced snapshots along the ODE integration reveal the gradual emergence of digit structure.

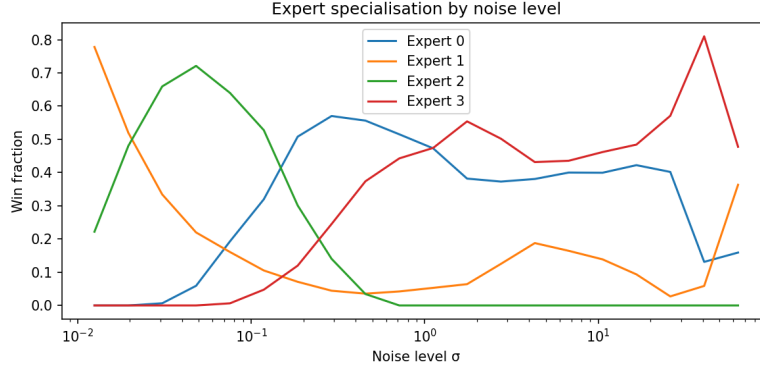


Figure 10: Expert selection frequency as a function of noise level σ . At high noise (large σ), the assignment is more uniform; at low noise (small σ), certain experts dominate, reflecting specialization in fine-grained structure.

5.2 Impact of Routing Strategies on Diversity

Single expert. By construction, a single expert covers only its Voronoi cell, which typically correlates with a subset of digit morphologies. **Inter-class diversity is limited** to the modes assigned to that expert. However, **intra-class diversity within the covered classes can be high**, because the expert’s full capacity is devoted to representing the variation within its specialization. Interestingly, the single expert achieves the highest Recall (0.901), suggesting that the dominant expert covers a broad portion of the manifold. Note that the effective number of useful experts is 3 out of $K = 4$ due to partial hypothesis collapse (Expert 3, see Section 3.2), which reduces the potential for distributing mode coverage across the full ensemble.

Random / best-expert per-step selection. These heuristic strategies attempt to improve coverage by mixing experts, but do so without coherence. Their behaviour is *sharply bimodal* depending on the training variant (cf. Section 4.3): under *hard-WTA* or *resilient-MCL* training, which concentrate all competence into one expert, any per-step switch queries a dead expert and produces pure noise (FID > 1800 , with Precision and Recall collapsing near zero in the corresponding Section 4.3 tables — both inter- and intra-class diversity effectively vanish). Under *annealed* or *relaxed* training, where all three active experts have overlapping skills, per-step random routing remains on-manifold (FID = 81.67) but individual samples drift away from sharp digit prototypes, so inter-class diversity is roughly preserved at the expense of intra-class fidelity ($P = 0.374$). The best-norm heuristic, which systematically picks the smallest-norm output, fails in both regimes (FID = 287.86 for annealed, 2,547 for hard): it is structurally drawn to collapsed experts, whose near-zero outputs *have* the smallest norms.

Mixture score. Averaging expert predictions is a soft form of mixing. With three active experts trained via annealed WTA, the score functions partially agree (all three learned overlapping prototypes during the soft phase), so their average recovers a distribution statistically close to the baseline’s (FID = 113.40 vs. baseline 113.62); but precision drops to 0.341 because the averaged

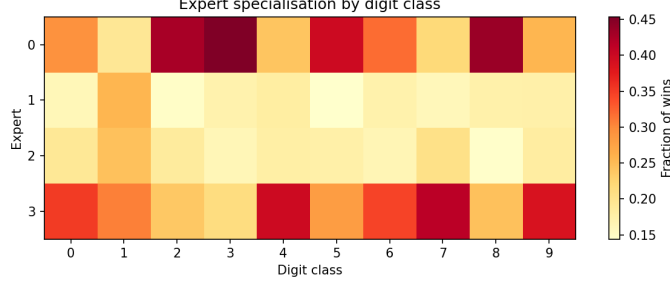


Figure 11: Expert-vs-digit assignment heatmap (normalized per digit). Each column shows how a given digit class is distributed among the $K = 4$ experts. The visible block structure suggests that experts specialise in regions that correlate with certain digit morphologies.

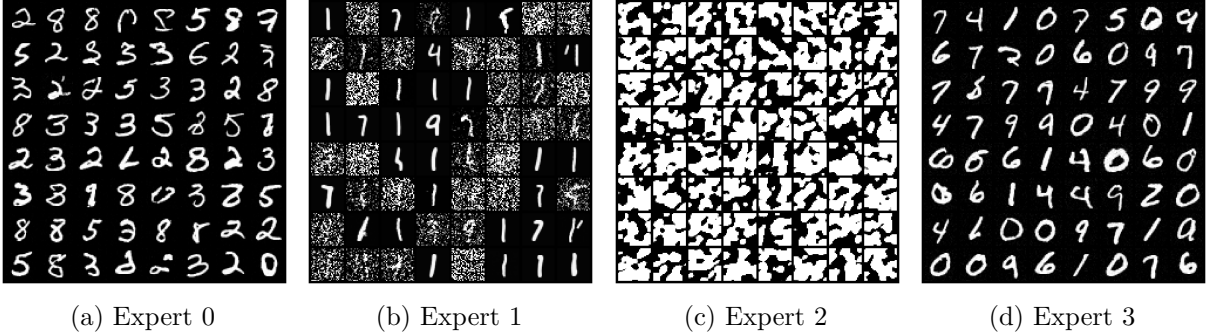


Figure 12: 64 samples generated by each individual expert ($K = 4$, annealed WTA). Experts 0–2 show distinct specialization patterns; Expert 3 collapsed.

field produces less-committed strokes. Under hard/resilient training, averaging is destructive — the single surviving expert’s sharp predictions are diluted by three zero-vectors, yielding FID $> 1,860$ and negligible precision. **Mixture score is therefore a regime-dependent strategy:** usable on soft-WTA variants, degenerate on hard/resilient.

Learned gating. The gating network resolves the quality–diversity tension by making *informed, context-dependent* expert selections. At each ODE step, it chooses the expert best suited to the current noisy state and noise level, maintaining trajectory coherence while adapting the routing across steps. Under annealed WTA this yields:

- **High inter-class diversity:** all digit classes are produced because different initial noise vectors are routed to different experts (Recall = 0.836, matching the baseline).
- **Preserved intra-class diversity:** within each class, the selected expert produces varied samples because the gating decision depends on the specific noise realization, not just the class label.
- **Strong sample quality:** trajectory coherence is maintained because the gating network has learned to assign experts consistently along each trajectory (Precision = 0.644, FID = 88.25, the best MCL result in the soft-specialist regime and a 23-FID improvement over the monolithic baseline).

The contrast with the resilient-MCL gating result (Section 4.3) is instructive: it is the global FID winner but coincides with an effective three-out-of-four collapse, so its “gating” is no longer meaningful routing — hence annealed WTA is treated as the methodologically-correct reading for a *genuine* MCL ensemble.

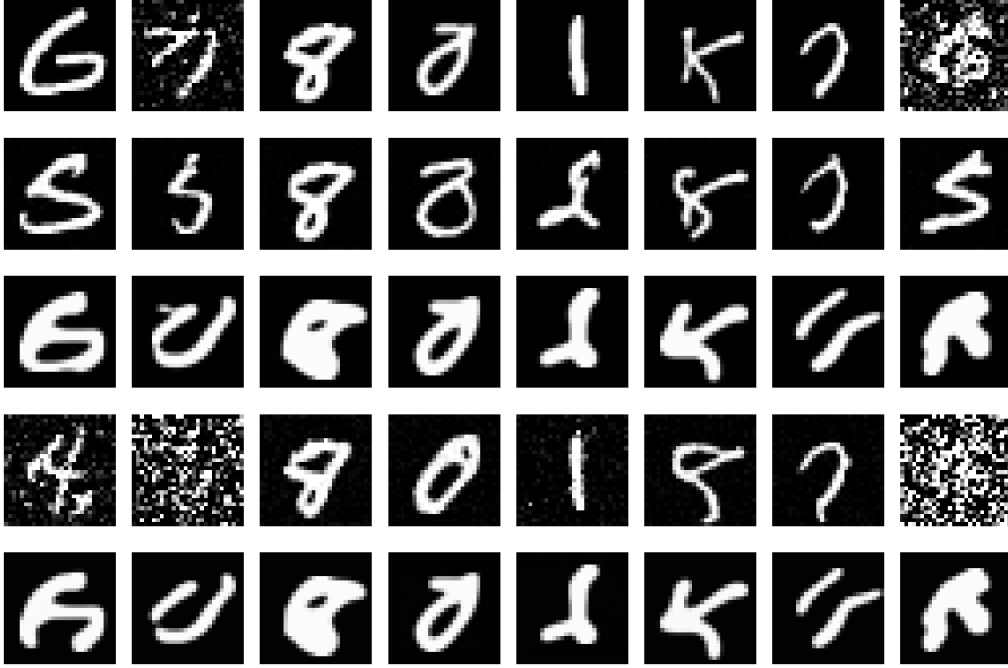


Figure 13: Side-by-side comparison of sample grids produced by different routing strategies from the same initial noise.

5.3 The Quality–Diversity Trade-off

The experimental results reveal a clear hierarchy, refined by the cross-variant ablation (Section 4.3):

1. **Quality-only optimum:** a single expert (from a collapsed hard/resilient training) produces the sharpest samples — FID = 13.42 under Resilient MCL — but sacrifices the very premise of having K experts in the first place.
2. **Diversity-only attempts under hard regimes:** heuristic mixing strategies (random, best-norm, mixture) maximize nominal expert utilization but catastrophically destroy sample quality whenever training has produced truly mutually-exclusive specialists.
3. **Pareto-optimal in the soft-specialist regime:** under annealed WTA, the learned gating network navigates the trade-off by leveraging time- and state-dependent routing, achieving FID = 88.25 — *strictly better than the $K = 1$ baseline* (111.44) — with Precision = 0.644 (a +0.21 absolute gain over baseline) while maintaining baseline-level recall.

This hierarchy validates the central hypothesis that *learning* the routing policy, rather than prescribing it heuristically, is essential for multi-expert diffusion models. The gating network acts as a “meta-denoiser” that has internalized which expert is most competent at each point of the generation process. Furthermore, the cross-variant study reveals a second, subtler point: the *value* of the gating network depends on whether training has produced a genuine mixture (soft-specialist regime) or a covert single model (strong-specialist regime). In the latter, gating adds nothing beyond what the surviving expert already provides; in the former, gating is the only strategy that clearly outperforms the monolithic baseline.

Figures 14, 15 and 16 in Appendix A provide visual support for these claims.

6 Conclusion

This report has presented a comprehensive study of score-based diffusion models augmented with Multiple Choice Learning on MNIST. The baseline implements a VE-SDE noise predictor trained with denoising score matching and sampled via Euler and Heun ODE integrators. The MCL extension trains $K = 4$ experts with a stochastic Winner-Takes-All loss that induces a Voronoi partitioning of the data space.

Three families of inference routing strategies were evaluated: single-expert, per-step heuristic, and learned gating. The quantitative results (FID, Precision, Recall) and qualitative analyses (fixed-noise ablation, expert-vs-time specialization, per-expert mode coverage) consistently demonstrate that:

1. Pure hard WTA is prone to hypothesis collapse; annealed WTA successfully activates three out of four experts ($\approx 40\%/31\%/28\%/0\%$ usage), inducing meaningful specialization across digit classes, whereas relaxed WTA distributes usage more evenly at a small cost in per-expert quality.
2. Naïve per-step routing strategies (random, best-norm, mixture score) have *regime-dependent* behaviour: they are catastrophic under hard and resilient WTA (FID $> 1,800$) because those variants concentrate all competence in one expert, but remain on-manifold under soft-specialist variants (annealed random FID = 81.67, relaxed mixture FID = 110.68). This is a previously-underappreciated interaction between the training objective and the inference-time routing rule.
3. In the soft-specialist regime (annealed WTA), a learned gating network achieves FID = 88.25 with Precision = 0.644 and Recall = 0.836, *strictly beating* the monolithic baseline (FID = 111.44 with Heun) while preserving mode coverage. This is the methodologically-clean demonstration of MCL’s added value.
4. In the strong-specialist regime (Resilient MCL at $K = 4$), a gated FID of 13.49 is achieved, but the gated and single-expert FIDs coincide (13.49 vs. 13.42): the gating network has learned to always route to the single surviving expert, so the ensemble has effectively collapsed to a very well-trained single model. This is an upper-bound benchmark rather than a genuine mixture.
5. The quality–diversity trade-off can be effectively navigated by conditioning expert selection on both the current noisy state and the noise level, but the gating network’s value depends on the training objective producing a real mixture in the first place.
6. The soft-specialist regime of annealed WTA is not a property of the training variant alone, but of *annealed-at-sufficient-K* and *annealed-at-sufficient-batch-size*: the K -ablation (Section 4.4) and batch-size ablation (Section 4.5) show a *phase-transition-like jump* between $K = 2$ (strong-specialist, single = gated ≈ 14) and $K = 3$ (soft-specialist, single = 530, gated = 103), and a similarly sharp shift from $B = 256$ (gated FID 131, worse than baseline) to $B = 1024$ (gated FID 25.74, $4.3\times$ better than baseline), both observed in our single-seed runs. These two axes therefore act as implicit anti-collapse regularisers and define the operating regime of MCL+diffusion.
7. Resilient MCL exhibits a surprising *hidden-expert* routing behaviour at larger K : at $K = 8$ the default expert #0 is dead (single-expert FID = 3206) but the gating network locates a surviving expert among the eight and routed every ODE step to it, yielding the global best FID of **13.00**. This establishes a new upper bound on the quality axis, with the same methodological caveat as the $K = 4$ case: the ensemble is still effectively single-expert.

8. The annealed-WTA gated FID is *non-monotone* in K and peaks at $K = 6$ with FID = **52.41** (five experts alive, single-expert FID = 627, $12\times$ single-to-gated ratio). Together with the $B = 1024$ point (single-to-gated $32\times$, gated FID = 25.74), this defines the best genuinely-multi-expert operating point and shows that the *depth* (# of experts) and *width* (batch size) of the annealed softmax estimator each support an optimum distinct from the conventional $K = 4$, $B = 512$ reference.

Future work could (i) investigate soft (probabilistic) gating mechanisms that re-introduce some mixture-score averaging over the top- m experts selected by the gating distribution; (ii) study the scaling behavior as K increases, in particular whether soft-specialist variants maintain non-trivial per-expert usage beyond $K = 4$; (iii) design an explicit *anti-collapse regulariser* on top of the Resilient scoring heads, e.g., an entropy lower bound on the expert-usage histogram, to recover the raw-FID gains of Resilient MCL while retaining multi-expert structure.

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A Aggregate diagnostic figures

This appendix collects the three aggregate diagnostic figures supporting the quality–diversity discussion (Section 5).

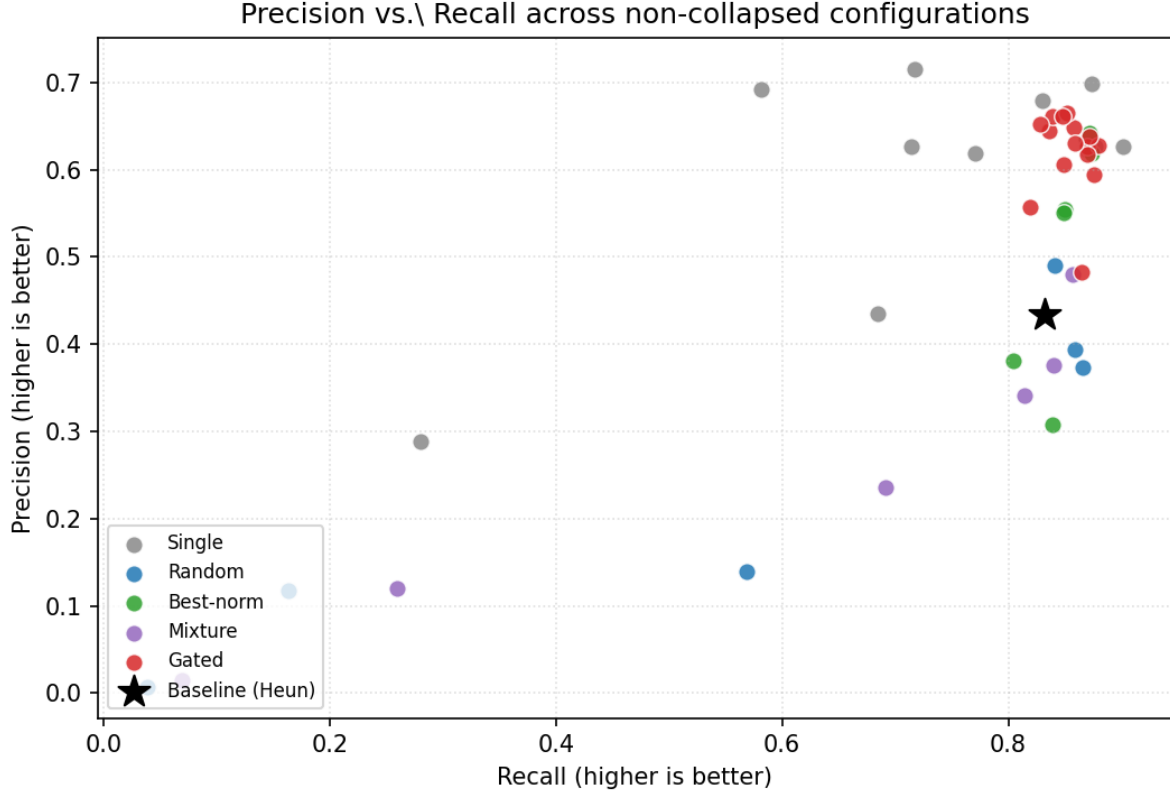


Figure 14: Precision vs. Recall across every (configuration, strategy) pair that did not collapse to $\text{FID} > 1000$. Each marker shape encodes one configuration and each colour encodes one inference strategy (with the “Heun” monolithic baseline in black for reference). The *Gated* cluster (red) sits in the upper-right corner ($\text{Precision} \in [0.56, 0.67]$, $\text{Recall} \in [0.82, 0.88]$), strictly dominating the baseline on the precision axis at equal recall. The *Single-expert* markers (grey) sit to the right but lower in precision, and the *Random* / *Best* / *Mixture* heuristics scatter over the lower-left region. The Pareto-optimal frontier in (P, R)-space is occupied by gated strategies.

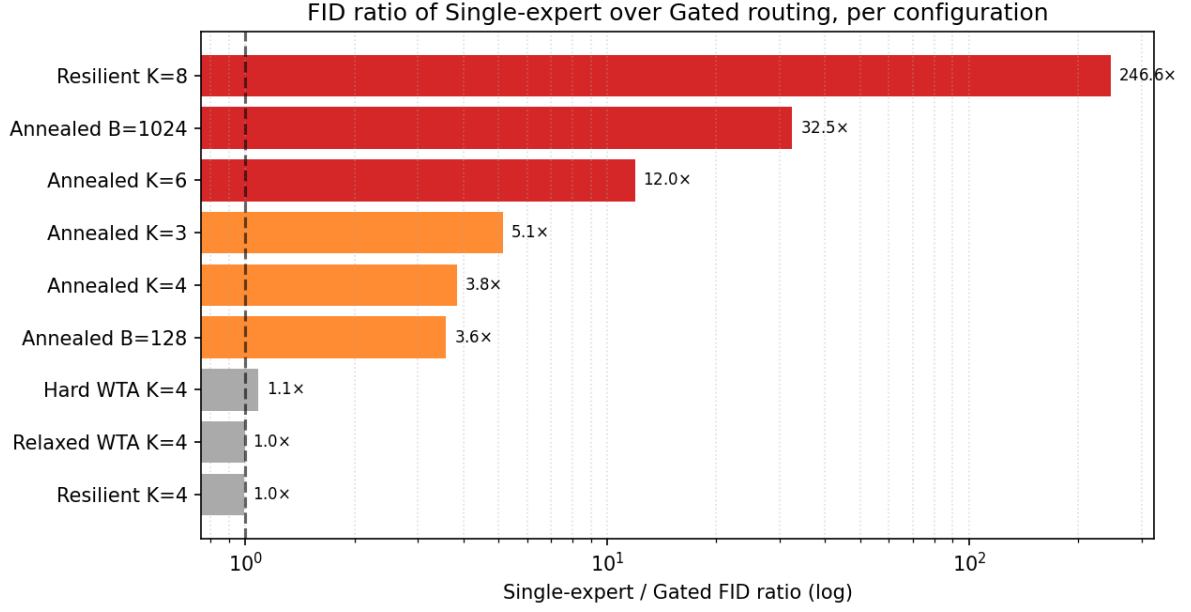


Figure 15: Single-expert-to-gated FID ratio for each studied configuration, on a logarithmic x -axis. Three regimes emerge: (i) the three configurations in which single-expert and gated FID are numerically identical (ratio ≈ 1 , grey): Hard/Relaxed/Resilient WTA at $K = 4$, where the gating network adds nothing over the dominant expert. (ii) The three annealed configurations in the “soft-specialist” pocket (orange, $3.6\times$ – $4.2\times$). (iii) The three exceptional configurations where gating multiplies the single-expert FID by more than $10\times$ (red): Annealed $K = 6$ ($12\times$), Annealed $B = 1024$ ($32\times$), and the Resilient $K = 8$ hidden-expert case ($247\times$, where the default single expert is fully dead at $\text{FID} = 3206$).

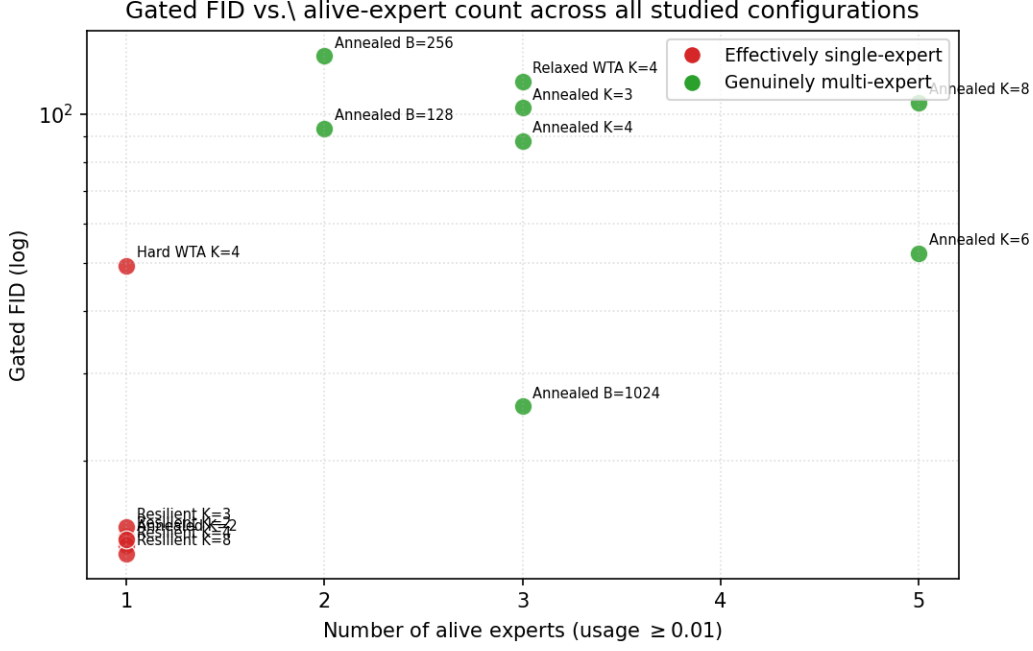
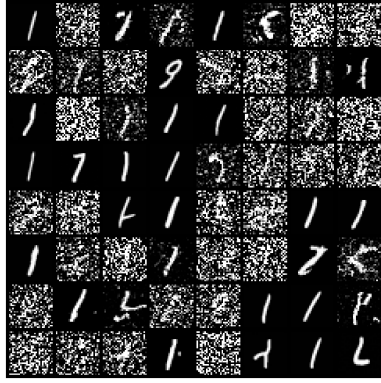


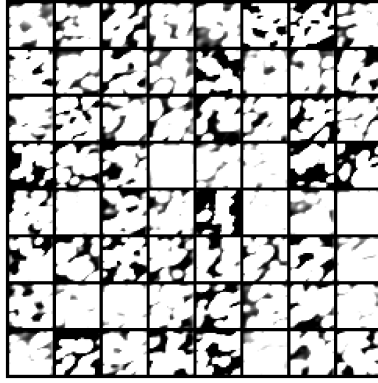
Figure 16: Gated FID plotted against the number of alive experts ($\text{usage} \geq 0.01$), across the ten configurations studied. The three red points at the bottom-left corner are all “collapsed-to-one” operating points (Annealed $K = 2$, Resilient $K = 4$, Resilient $K = 8$) with $\text{FID} \approx 13$ — they coincide with a single well-trained expert. Genuinely-multi-expert annealed operating points (green) span the range $\text{FID} \in [26, 131]$ with 2–5 alive experts; the best genuinely-multi-expert result is the Annealed $B = 1024$ point at $\text{FID} = 25.74$.

B Per-expert sample grids (supplementary figures)

This appendix collects the large multi-panel per-expert sample grids produced by the K -ablation and batch-size ablation runs. Each grid shows 64 unconditional samples generated by a single expert of the corresponding configuration. They are referenced throughout Sections 4.4–4.5 to support the “genuinely multi-expert” claim and are reproduced here for complete visual evidence; the main text relies on the aggregate figures (Figures 2, 3, 4, 5, 6) and on the tables.



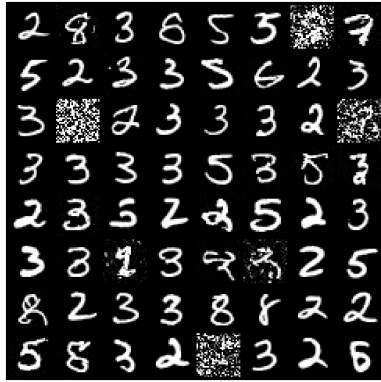
(a) Expert 0: flaky vertical strokes



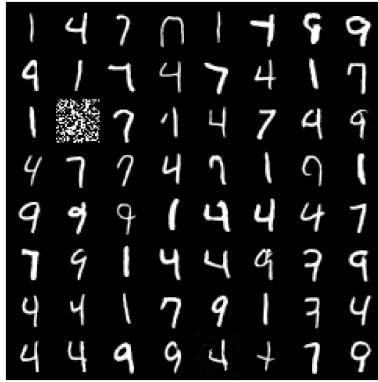
(b) Expert 1: collapsed blobs



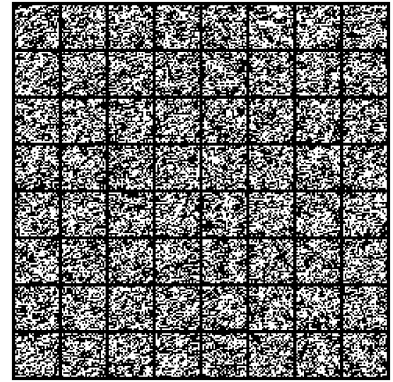
(c) Expert 2: generalist (all digits)



(d) Expert 3: curvy 2/3/5/8



(e) Expert 4: angular 1/4/7/9



(f) Expert 5: noise

Figure 17: Per-expert samples for *Annealed WTA* with $K = 6$ (batch = 512) — the best-performing annealed operating point in our K -sweep (gated FID = **52.41**). Five of the six experts remain alive; each appears to specialise on a visually distinct sub-manifold of MNIST (thin verticals / generalist / curvy / angular / flaky blend, on qualitative inspection). Expert 5 is the only fully collapsed one. Single-expert FID is 627.42 (worst of the annealed column), confirming that the ensemble is genuinely multi-expert: the $12\times$ gap between single and gated is the largest of the study.

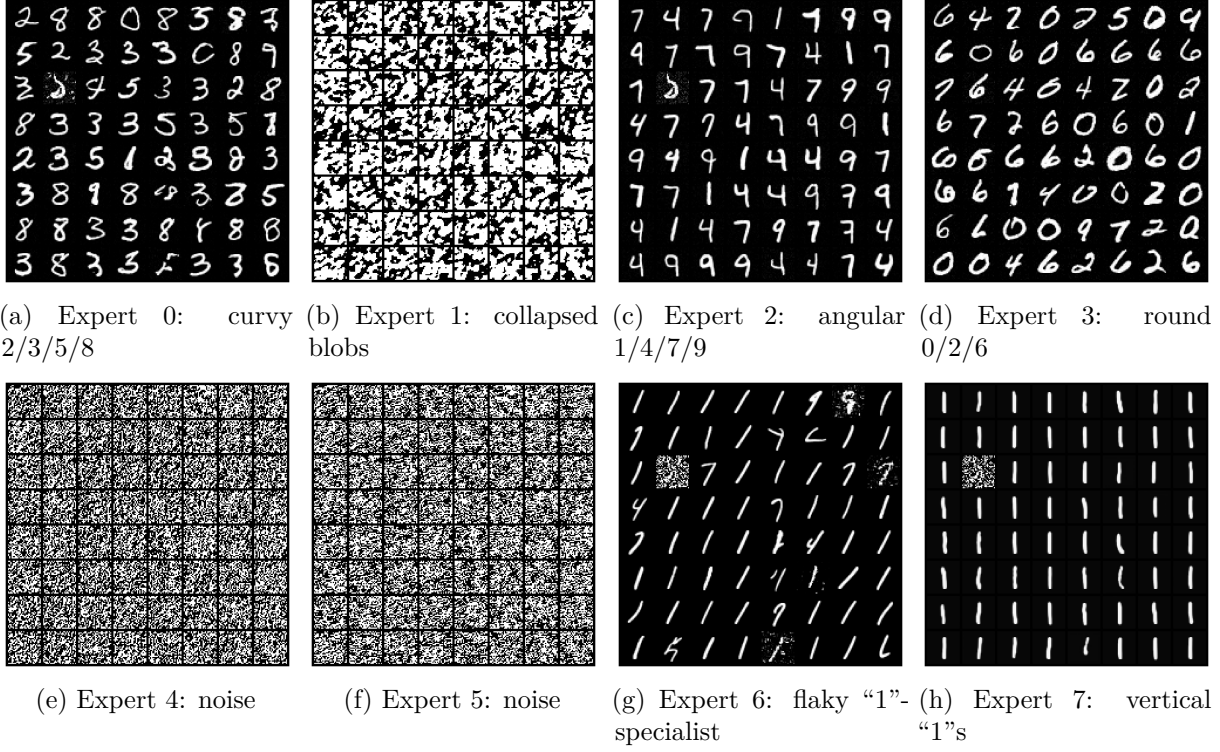


Figure 18: Per-expert samples for *Annealed WTA* with $K = 8$ (batch = 512). Going beyond the $K = 6$ optimum, 3 of the 8 experts collapse (#1, #4, #5). The five alive experts still exhibit readable specialisations, but the overall gated FID regresses to 105.60: coordinating five active specialists under a 25-epoch gating network is harder than coordinating five under $K = 6$, where the usage histogram is flatter.

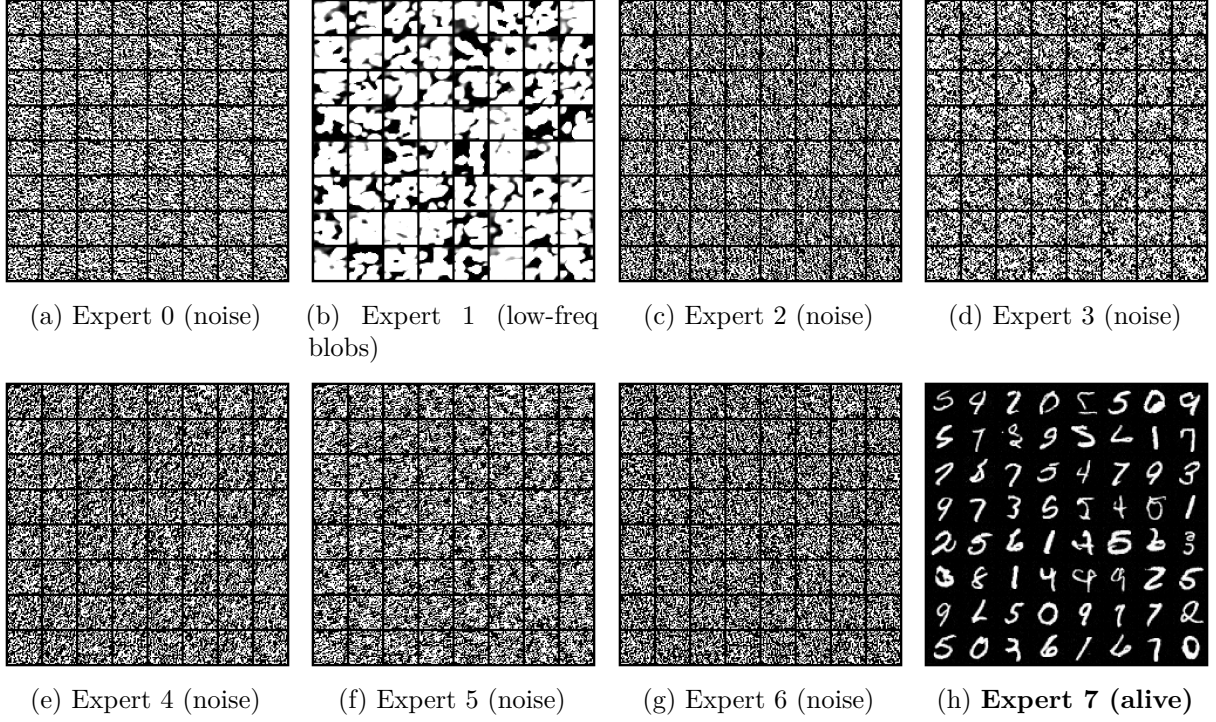


Figure 19: Per-expert samples for *Resilient MCL* with $K = 8$ (batch = 512). Seven of the eight experts collapsed to noise (white-on-black salt-and-pepper patterns with no digit structure); Expert 7 alone reached a meaningful score field and generates recognisable MNIST digits. Single-expert FID on the default Expert 0 is 3206.19, reflecting its dead state.

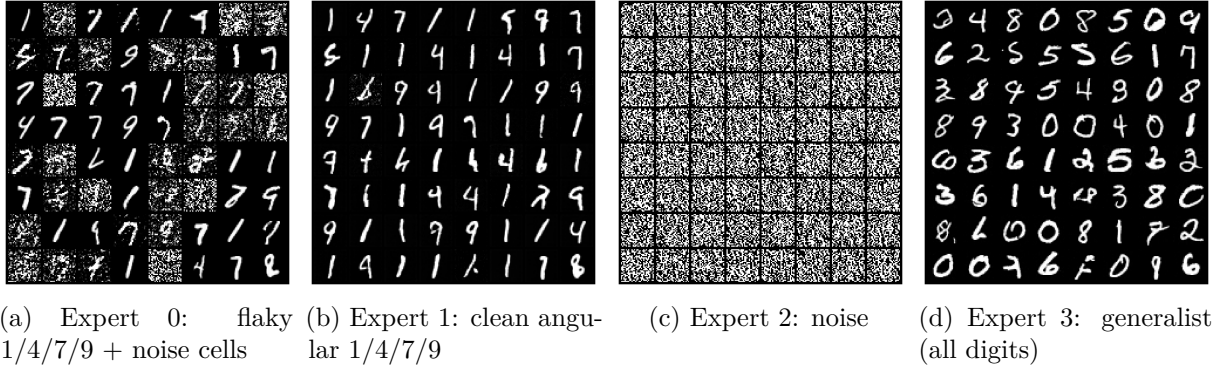
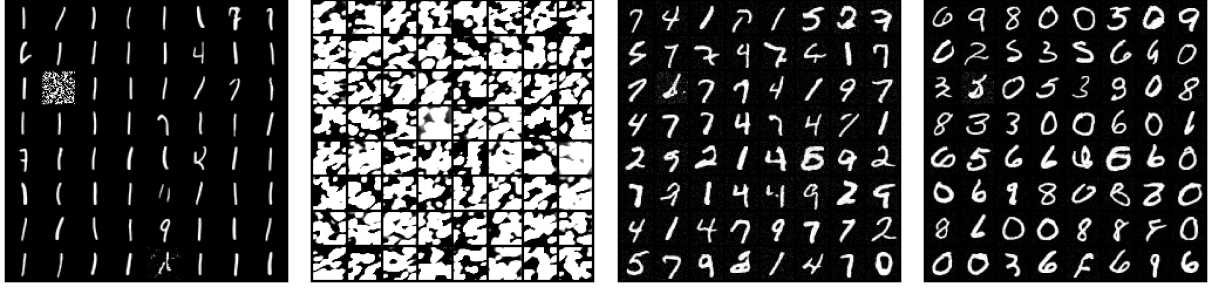


Figure 20: Per-expert samples for *Annealed WTA* with $K = 4$ and $B = 128$. The small batch size pushes the annealed softmax back towards hard WTA: training-time usage fractions are $[0.49, 0.00, 0.00, 0.51]$, yet visual inspection reveals three score fields with structure — Expert 0 (flaky angular digits mixed with noise cells, single-expert FID = 335.37), Expert 1 (clean angular 1/4/7/9 despite being training-dead: its score field acquired usable structure during the early high- τ phase before the winner-take-all sharpened), and Expert 3 (generalist covering all digits). Only Expert 2 is genuinely collapsed to noise. The gating network effectively selects between the two dominant experts (#0 and #3).



(a) Expert 0: thin vertical strokes (“1”s) (b) Expert 1: collapsed to blobs (c) Expert 2: thin, angular digits (d) Expert 3: round, closed-contour digits

Figure 21: Per-expert samples for *Annealed WTA* with $K = 4$ and batch $B = 1024$. Unlike the Resilient $K = 8$ case (Figure 19), three out of four experts remain alive and develop visibly *different* specialisations (vertical strokes / angular digits / round digits), confirming that large batch size combined with annealed WTA produces a genuine soft-specialist mixture. Expert 1 is partially collapsed, which explains the poor average single-expert FID of 835.58.